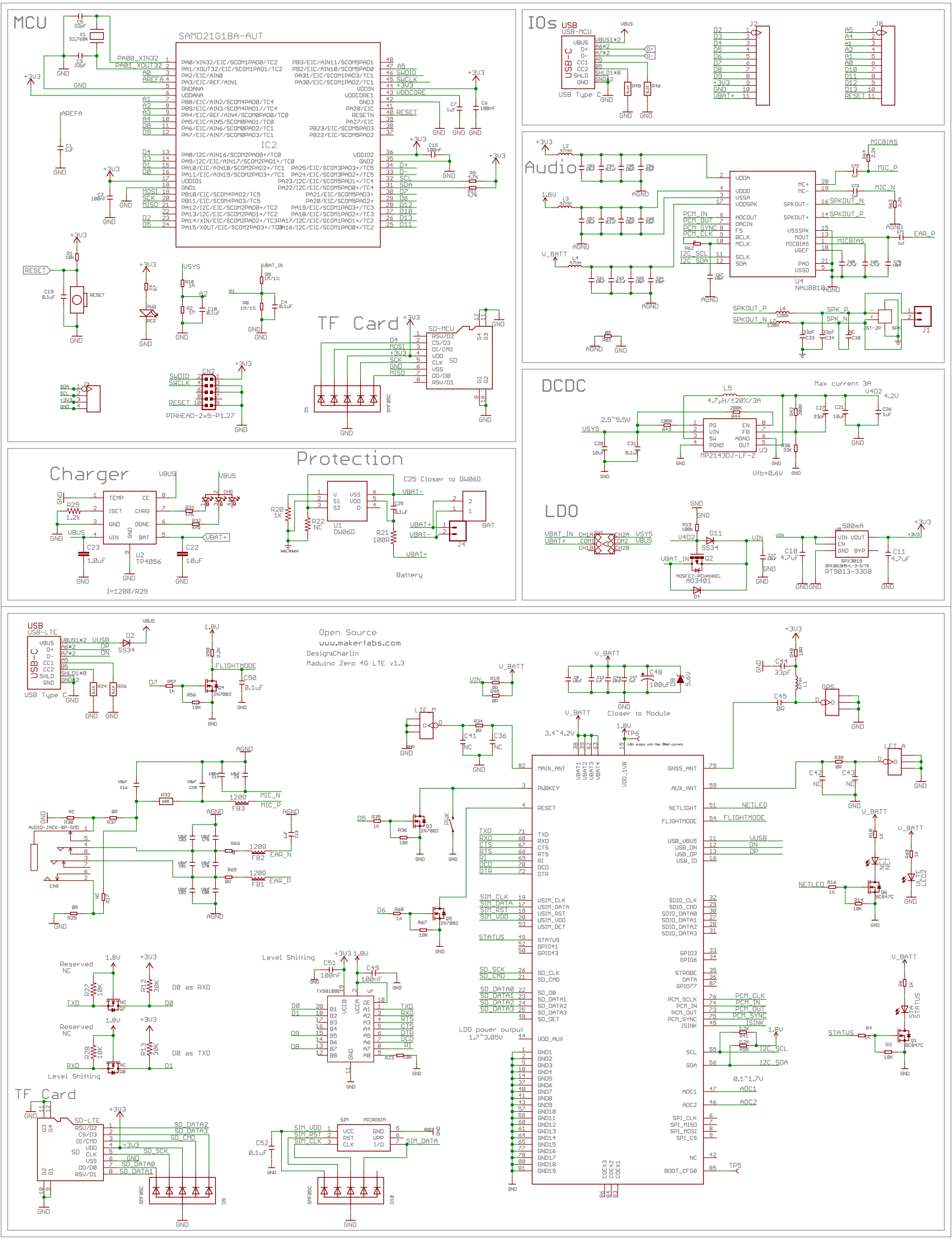
MCU
SAMD21G18A-AUT
IC2
X1 32.768K
C9 22pF
C3 22pF
AREFA
C1 1uF
C2 100nF
C7 1uF
C6 100nF
C15 100nF
R9 4.7K
R1 10K
RESET
C19 0.1uF
LED1 RED
VSYS
R10 1M
R2 1M
C10 0.1uF
VBAT_IN
R8 1M/1%
R9 1M/1%
C4 0.1uF
CN2
SWDIO
SWCLK
RESET
PINHEAD-2x5-P1.27
J3
SDA
SCL
+3V3
GND
TF Card
SD-MCU
RSV/D2
CS/D3
DI/CMD
VDD
CLK
VSS
DO/D0
RSV/D1
SD
D4
MOSI
SCK
MISO
SMF05C
D1
D9
IOs
USB
USB-MCU
USB-LTE
USB Type C
VBUS
VBUS1*2
A6*2
A7*2
A5
B5
SHLD1*B
GND*2
R48 5.1K
R46 5.1K
J2
J8
Audio
L2 47nH
L3 120nH
L4 47nH
V_BATT
1.8V
AGND
MICBIAS
MIC_P
MIC_N
SPKOUT_N
SPKOUT_P
EAR_P
PCM_IN
PCM_OUT
PCM_SYNC
PCM_CLK
I2C_SCL
I2C_SDA
VDDA
VDDD
VSSA
VDDSPK
ADCOUT
DACIN
FS
BCLK
MCLK
SCLK
SDA
MC+
MC-
SPKOUT-
SPKOUT+
VSSSPK
MOUT
MICBIAS
VREF
PAD
VSSD
U4
NAU8810
SPK_P
SPK_N
J1
DCDC
Max current 3A
L5
4.7μH/±20%/3A
V4D2
4.2V
2.5~5.5V
VSYS
MP2143DJ-LF-Z
U3
Vfb=0.6V
LDO
VBAT_IN
VBAT+
CH1A
CH1B
CH2A
CH2B
COM1
COM2
VSYS
VBUS
D11
SS34
Q2
MOSFET-DCHANNEL
AO3401
D4
VIN
500mA
VIN VOUT
EN
GND BYP
SPX3819
RT9013-33GB
C10 4.7uF
C11 4.7uF
Charger
TEMP
CE
ISET
CHRG
GND
DONE
VIN
BAT
U2
TP4056
R29 1.2K
R31 470
R32 470
CH6
C23 1.0uF
C22 10uF
I=1200/R29
Protection
C25 Closer to DW06D
U1
DW06D
R20 1K
R22 NC
R21 100R
J4
BAT
Battery
Open Source
www.makerfabs.com
Design:Charlin
Maduino Zero 4G LTE v1.3
FLIGHTMODE
Q4 2N7002
C50 0.1uF
D7
R57 1K
R56 10K
V_BATT
VIN
R49 0R
C48 100uF
5.5V
Closer to Module
3.4~4.2V
1.8V
TP6
1.8V output with Max 50mA current
LTE_M
R34 0R
C41 NC
C36 NC
GPS
C44 33pF
C45 0R
L1 47nH
C49 10R
LTE_A
C42 NC
C43 NC
R39 0R
MIC_N
MIC_P
FB3 120Ω
EAR_N
FB2 120Ω
EAR_P
FB1 120Ω
AUDIO-JACK-8P-SMD
EAR
D5 R35 1K
Q3 2N7002
PWK
D6 R68 1K
Q5 2N7002
R67 10K
MAIN_ANT
PWRKEY
RESET
TXD
RXD
CTS
RTS
RI
DCD
DTR
SIM_CLK
SIM_DATA
SIM_RST
SIM_VDD
STATUS
SD_SCK
SD_CMD
SD_DATA0
SD_DATA1
SD_DATA2
SD_DATA3
USIM_CLK
USIM_DATA
USIM_RST
USIM_VDD
USIM_DET
GPIO41
GPIO43
SD_CLK
SD_D0
VDD_1V8
VDD_AUX
LDO power output
1.7~3.05V
GND1
GND2
GND3
GND4
GND5
GND6
GND7
GND8
GND9
GND10
GND11
GND12
GND13
GND14
GND15
GND16
GND17
GND18
GND19
GNSS_ANT
AUX_ANT
NETLIGHT
FLIGHTMODE
USB_VBUS
USB_DN
USB_DP
USB_ID
VUSB
DN
DP
SDIO_CLK
SDIO_CMD
SDIO_DATA0
SDIO_DATA1
SDIO_DATA2
SDIO_DATA3
GPIO3
GPIO6
STROBE
DATA
GPIO77
PCM_CLK
PCM_IN
PCM_OUT
PCM_SYNC
ISINK
SCL
SDA
I2C_SCL
I2C_SDA
0.1~1.7V
ADC1
ADC2
SPI_CLK
SPI_MISO
SPI_MOSI
SPI_CS
NC
BOOT_CFG0
TP5
NETLED
LTE LED2
Q6 BC847C
R14 10K
R16 1K
R18 1K
R48 1K
STATUS
R4 1K
Q1 BC847C
R3 10K
R41 2.2K
R55 2.2K
Reserved
NC
R27 10K
R12 30K
D0 as RXD
TXD
D0
Q7
R28 10K
R13 30K
D0 as TXD
RXD
D1
Q8
Level Shifting
C51 100nF
C49 100nF
TXS0108E
U7
VCCB
VCCA
OE
D0
D1
D9
D8
TXD
RXD
RTS
CTS
DTR
DCD
RI
R23 10K
TF Card
SD-LTE
RSV/D2
CS/D3
DI/CMD
VDD
CLK
VSS
DO/D0
RSV/D1
SD_DATA2
SD_DATA3
SD_CMD
SD_SCK
SD_DATA0
SD_DATA1
SIM
MICROSIM
VCC
RST
CLK
GND
VPP
I/O
SIM_VDD
SIM_RST
SIM_CLK
SIM_DATA
C52 0.1uF
D10